

Brock Sorenson
CSE 412
Assignment 04

Code Repository:

The code is available at the following Github Repository:
<https://github.com/btsorens/CSE412Assignment04>

Schema:

An overview of the schema created in schema.sql

Players Table

```
CREATE TABLE players (  
    player_id SERIAL PRIMARY KEY,  
    first_name VARCHAR(50) NOT NULL,  
    last_name VARCHAR(50) NOT NULL,  
    date_of_birth DATE NOT NULL  
        CHECK (date_of_birth BETWEEN '1970-01-01' AND '2010-12-31'),  
    hometown VARCHAR(100),  
    state VARCHAR(2),  
    position VARCHAR(10),  
    height INTEGER CHECK (height BETWEEN 68 and 80),  
    weight INTEGER CHECK (weight BETWEEN 150 and 300),  
    jersey_number INTEGER CHECK (jersey_number BETWEEN 0 and 99)  
);
```

Gamestats Table

```
CREATE TABLE gamestats (  
    stat_id SERIAL PRIMARY KEY,  
    player_id INTEGER REFERENCES players(player_id) ON DELETE CASCADE,  
    date_of_game DATE NOT NULL,  
    stadium_name VARCHAR(100),  
    rush_yards INTEGER DEFAULT 0,  
    num_rushes INTEGER DEFAULT 0,  
    rec_yards INTEGER DEFAULT 0,  
    num_receptions INTEGER DEFAULT 0,  
    points INTEGER DEFAULT 0,  
    snaps_played INTEGER CHECK (snaps_played >= 0)  
);
```

Queries:

On the web interface the user can choose between Query Type: “Players (Single Table)” and “Players and Gamestats (Joined Tables)” which runs these queries on the backend.

Single Table Query on Players Table By Last Name

```
SELECT player_id, first_name, last_name, date_of_birth,  
hometown, state, position,height, weight, jersey_number  
FROM players  
WHERE last_name LIKE <searched last name>  
ORDER BY last_name ASC, first_name ASC;
```

Joined Table Query on Players and Gamestats by Last Name

```
SELECT p.first_name, p.last_name, gs.date_of_game, gs.stadium_name,  
       gs.rush_yards, gs.num_rushes, gs.rec_yards, gs.num_receptions,  
       gs.points, gs.snaps_played  
FROM players p  
JOIN gamestats gs  
ON p.player_id = gs.player_id  
WHERE p.last_name LIKE <searched last name>  
ORDER BY p.last_name ASC, p.first_name ASC;
```

Indices:

On the web interface the user can choose to run the query with Query Mode: “Non-Indexed” or “Indexed”, which creates or removes the following indices on the backend before executing the query.

```
CREATE INDEX IF NOT EXISTS last_name_index ON players(last_name);  
CREATE INDEX IF NOT EXISTS player_id_index ON gamestats(player_id);
```

Web Interface Screenshots:

Non-Indexed Single Table Query on Players

Search Players

Query Mode: Query Type:

Search Results:

Andrew Pitts (ID: 1984)
Jersey Number:48 | Position: TE | Hometown: Millermouth, TX
DOB:1989-12-31 | Height (in):69 | Weight (lbs):221

Angela Pitts (ID: 8414)
Jersey Number:82 | Position: RB | Hometown: Andrewport, MN
DOB:1999-08-28 | Height (in):78 | Weight (lbs):192

Carlos Pitts (ID: 7275)
Jersey Number:21 | Position: TE | Hometown: Laurentown, TX
DOB:1996-05-21 | Height (in):79 | Weight (lbs):241

Gregory Pitts (ID: 1766)
Jersey Number:97 | Position: RB | Hometown: Port Heathershire, NC
DOB:2008-03-04 | Height (in):78 | Weight (lbs):164

James Pitts (ID: 8741)
Jersey Number:80 | Position: WR | Hometown: Laurenton, DC
DOB:1990-08-28 | Height (in):71 | Weight (lbs):166

Query execution time 2.96 ms

Indexed Single Table Query on Players

Search Players

Pitts

Query Mode: Indexed

Query Type: Players (Single Table)

Search

Search Results:

Andrew Pitts (ID: 1984)
Jersey Number:48 | Position: TE | Hometown: Millermouth, TX
DOB:1989-12-31 | Height (in):69 | Weight (lbs):221

Angela Pitts (ID: 8414)
Jersey Number:82 | Position: RB | Hometown: Andrewport, MN
DOB:1999-08-28 | Height (in):78 | Weight (lbs):192

Carlos Pitts (ID: 7275)
Jersey Number:21 | Position: TE | Hometown: Laurentown, TX
DOB:1996-05-21 | Height (in):79 | Weight (lbs):241

Gregory Pitts (ID: 1766)
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DOB:2008-03-04 | Height (in):78 | Weight (lbs):164

James Pitts (ID: 8741)
Jersey Number:80 | Position: WR | Hometown: Laurenton, DC
DOB:1990-08-28 | Height (in):71 | Weight (lbs):166

Query execution time 1.85 ms

Non-Indexed Joined Table Query on Players and Gamestats

Search Players

Pitts

Query Mode: Non-Indexed

Query Type: Players and Game Stats (Joined Tables)

Search

Search Results:

Angela Pitts
Game Date: 2025-04-14 | Stadium: Holder LLC Stadium | Points: 0
Rush Yards: 29 | Rushes: 3
Receiving Yards: 19 | Receptions: 2
Snaps Played: 38

Carlos Pitts
Game Date: 2026-04-06 | Stadium: Parker-Santiago Stadium | Points: 7
Rush Yards: 129 | Rushes: 19
Receiving Yards: 54 | Receptions: 14
Snaps Played: 63

Carlos Pitts
Game Date: 2020-10-26 | Stadium: King Inc Stadium | Points: 3
Rush Yards: 31 | Rushes: 9
Receiving Yards: 136 | Receptions: 2
Snaps Played: 19

Carlos Pitts
Game Date: 2025-11-18 | Stadium: Coleman, Gray and Obrien Stadium | Points: 7
Rush Yards: 50 | Rushes: 7
Receiving Yards: 14 | Receptions: 7
Snaps Played: 38

Gregory Pitts
Game Date: 2021-05-12 | Stadium: Murphy and Sons Stadium | Points: 6
Rush Yards: 15 | Rushes: 7
Receiving Yards: 88 | Receptions: 7
Snaps Played: 53

Query execution time 3.69 ms

Indexed Joined Table Query on Players and Gamestats

Search Players

Pitts

Query Mode: Indexed

Query Type: Players and Game Stats (Joined Tables)

Search

Search Results:

Angela Pitts

Game Date: 2025-04-14 | Stadium: Holder LLC Stadium | Points: 0

Rush Yards: 29 | Rushes: 3

Receiving Yards: 19 | Receptions: 2

Snaps Played: 38

Carlos Pitts

Game Date: 2025-11-18 | Stadium: Coleman, Gray and Obrien Stadium | Points: 7

Rush Yards: 50 | Rushes: 7

Receiving Yards: 14 | Receptions: 7

Snaps Played: 38

Carlos Pitts

Game Date: 2020-10-26 | Stadium: King Inc Stadium | Points: 3

Rush Yards: 31 | Rushes: 9

Receiving Yards: 136 | Receptions: 2

Snaps Played: 19

Carlos Pitts

Game Date: 2026-04-06 | Stadium: Parker-Santiago Stadium | Points: 7

Rush Yards: 129 | Rushes: 19

Receiving Yards: 54 | Receptions: 14

Snaps Played: 63

Gregory Pitts

Game Date: 2021-05-12 | Stadium: Murphy and Sons Stadium | Points: 6

Rush Yards: 15 | Rushes: 7

Receiving Yards: 88 | Receptions: 7

Snaps Played: 53

Query execution time 2.72 ms

Summary Analysis:

The indexed search performed much better in both the single table query and the joined table query. In the single table query without the index the execution time was 2.96 milliseconds. After rerunning the query with an index on last_name, the query execution time was 1.85 milliseconds. This was an improvement of over 1 millisecond and 37.5% quicker.

The results were significant on the joined query as well. The non-indexed query took 3.69 milliseconds of execution time. The indexed query took 2.72 milliseconds, about 1 millisecond quicker and a 26.3% efficiency improvement.